

David Deutsch also points out that our imaginations, which seek to model the world, mimic reality as virtual reality generators: we model environments and imagine how they could play out in different branches.

3.1 Reading as a multiversal act

When a piece of literature is read, the underlying multiverse shapes the reader’s interpretations and expectations. The structure which determines the meaning of a piece as experienced by a reader is not the linear-time story but the implicit, counterfactual past/present/future plexus surrounding each point in the text given by the reader’s projective and interpretive imagination.

More concretely stated, at each moment in a story, there is uncertainty about how dynamics will play out (will the hero think of a way out of their dilemma?) as well as uncertainty about the hidden state of the present (is the mysterious mentor good or evil?). Each world in the superposition not only exerts an independent effect on the reader’s imagination but interacts with counterfactuals (the hero is aware of the uncertainty of their mentor’s moral alignment, and this influences their actions).

The reader simulates the minds of the characters and experiences the multiverses evoked by the story.

3.2 Writing as a multiversal act

A writer may have a predetermined interpretation and future in mind or may write as a means of exploring the interpretative and/or dynamic multiverse. Regardless, a writer must be aware of the multiplicity which defines the readers’ and characters’ subjective experiences as the shaper of the meaning and dynamics of the work. The writer thus seeks to simulate and manipulate that multiplicity.

We propose that generative language models in their multiversal modality can serve as an augmentation to and be augmented by the writer’s inherently multiversal imagination.

3.3 Writing multiverses

So far we’ve implicitly assumed that, despite the multiversal forces at work, the writer’s objective is to eventually compose a single history. However, language models naturally encourage writing explicitly multiversal works.

In the same way that hypertext transcended the limitations the linear order in which physical books are read, exciting a surge of multiversal fiction [2], language models introduce new possibilities for *writing* nonlinear narratives.

After all, it’s only a small leap from incorporating multiverses in the brainstorming process to including them in the narrative. Counterfactual branches often occur in traditional fiction in the form of imaginary constructs, and our minds are naturally drawn to their infinite possibility [1].

4 Interfaces

We propose the creation of new tools to allow writers to work alongside language models to explore and be inspired by the multiverses already hiding in their writing.

Research into hypertext writing tools has been ongoing for more than two decades and has produced notable tools like StorySpace[3]. However, the issue of hypertext interfaces assisted by language models is a newer development, as only very recently have language models become advanced enough to be useful in the writing process [9]. Likewise, there has been significant research into interactive narratives, including in branching, multiversal settings [10, 11], but never one in which the human and the language assistant can act together as such high-bandwidth partners.

As has been shown in past hypertext interface design studies [8], the primary concern in the creation of an interface for writing multiverse story is the massive amount of information that could be shown to the writer. If intuitive user experience is not central to the design of the program, this information will feel overwhelming and functionally prevent the user from leveraging the power offered by multiverse access at all.

An effective multiversal interface should allow the writer, with the aid of a generative language model, to expose, explore, and exploit the interpretational and dynamic multiplicity of a passage. Not only will such a tool allow the user to explore the ways in which a scenario might play out, such an interface will also expose previously unnoticed ambiguities in the text (and their consequences).

Depending on the design of the interface and the way the user approaches it, many different human-AI collaborative workflows are possible. Ideally, the interface should give the user a sense of creative superpowers, providing endless inspiration combined with executive control over the narrative, as well as allowing and encouraging the user to intervene to any degree.

4.1 Progress so far

Over the past several months, we have prototyped and tested several iterations of multiversal writing tools using GPT-3 as the generation function.

The demand for a multiversal writing application grew from use of GPT-3 as a more standard linear writing assistant. It became increasingly clear, as users sought greater interaction bandwidth and more efficient ways to structure and leverage the model’s ideas, that an interface which organizes the model’s outputs in a branching tree would be more effective.

The early results we have seen leave no doubt about the power of language models as writing assistants. Our small cohort of five beta users have, alongside GPT-3, co-written linear and nonlinear stories spanning the equivalent of thousands of pages of often astonishing ingenuity and beauty and surprisingly long-range coherence. Three users have reported a sense of previously unimagined creative freedom and expressive power.

However, it has also become evident that much more research and development is necessary. In our beta-tests, we’ve found that flaws in interface design can easily overwhelm or damage a feeling of ownership over the work produced. Below we will share some of our findings, which represent only the first step in creating a true interface between the creative mind and the machine.

4.2 Multiple visualizations

We have found that a visual representation of the branching structure of the narrative helps users conceptualize and navigate fractal narratives. This view (called **visualize**) displays the flow of pasts and futures surrounding each node (Figure 3) and zooming out displays the global structure of the multiverse (Figure 4). The **visualize** view allows users to expand and collapse nodes and subtrees, as well as “hoist” any node so that it acts as the root of the tree. Altering the topology of the tree, (e.g. reassigning children to different parents, splitting and merging nodes) is more intuitive for users in the **visualize** view than the linear view.

In addition to tree-based multiverse visualization, the **read** view displays the text of a node and its ancestry in a single-history format (Figure 5).

4.3 Multiverse navigation

With a generative language model, story multiverses can quickly become too large to navigate through node connections alone. To assist navigation, we have implemented the following features:

- **Search** all text or text in a subtree and/or text in a node’s ancestry

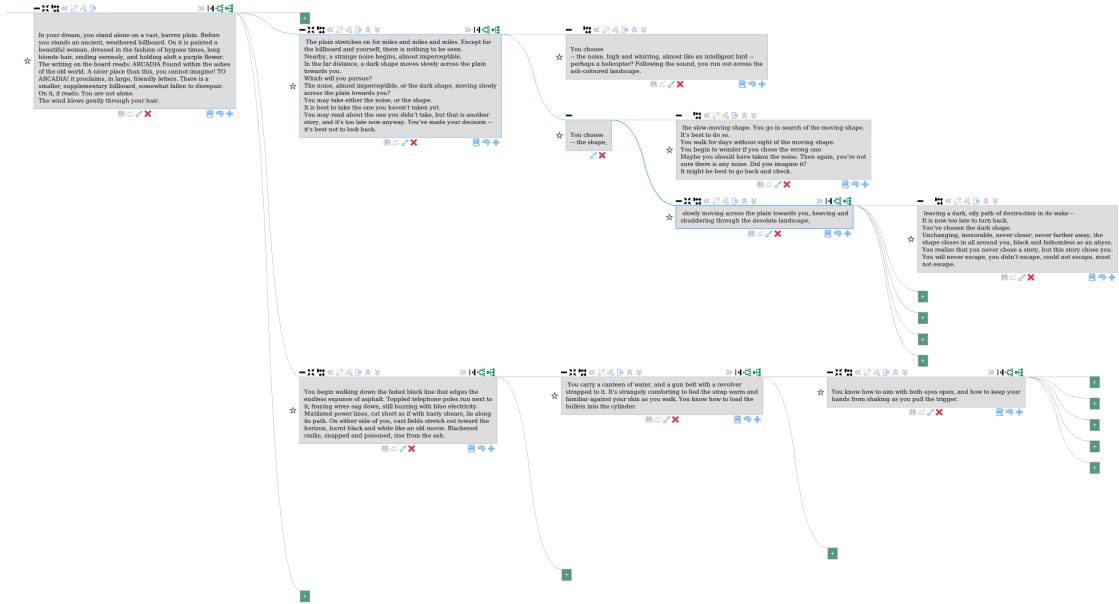


Figure 3: Visualize view

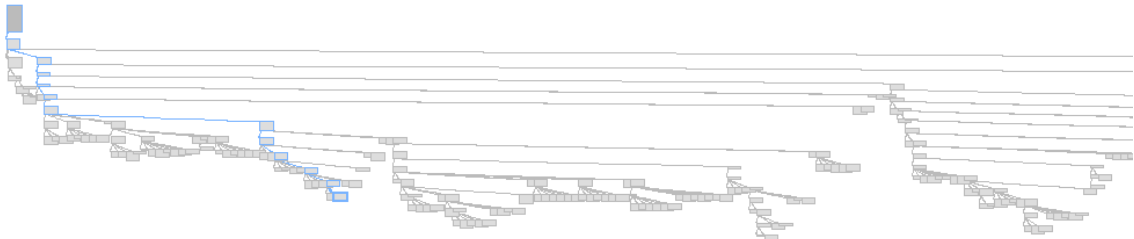


Figure 4: Zoomed-out visualization of a nonlinear story